

Power Moment Identities on Weight Distributions in Error Correcting Codes

VERA PLESS

Air Force Cambridge Research Laboratories, Bedford, Massachusetts

A series of identities relating the weight distribution in any code space to the weight distribution in the orthogonal code space has been given by Mrs. J. MacWilliams (1962). Here we derive a series of power moment identities from the MacWilliams identities. An earlier result of Assmus and Mattson is shown to be equivalent to the third power moment identity.

A unique solution to the power moment identities is given under certain conditions. Applications are given.

I. INTRODUCTION

A. SUMMARY

A series of identities relating the weight distribution in any code space to the weight distribution in the orthogonal code space has been given by Mrs. J. MacWilliams (1962). Here we derive a series of power moment identities from the MacWilliams identities. An earlier result of Assmus and Mattson (1961) is shown to be equivalent to one of these identities. For $q = 2$, their result was also proven by Zierler (1962).

We study states of partial information on weight distributions such that the power moment identities determine uniquely the remaining values. Applications are made in determining the weight distributions of certain cyclic codes.

B. DEFINITIONS

These are the definitions and notations relevant primarily to Section II.

Let V be a vector space of dimension n over $GF(q)$, the Galois field of q elements, q a prime power. Let A be a subspace of V , of dimension k . Two vectors $v = (v_0, v_1, \dots, v_{n-1})$ and $w = (w_0, w_1, \dots, w_{n-1})$ in V are said to be orthogonal if $\sum_{i=0}^{n-1} v_i w_i$ is 0 in $GF(q)$. Let B be the set of all vectors in V orthogonal to each vector in A . B is called the orthogonal of A .

If $v = (v_0, \dots, v_{n-1})$ is a vector in V , the weight of v is defined to be the number of nonzero v_i . A_i will denote the number of vectors of weight i in A and B_i will denote the number of vectors in weight i in B .

The following are the facts relevant to Section III.

We recall that a code space is called cyclic if it is invariant under the coordinate permutation corresponding to the mapping $i \rightarrow i + 1$ (modulo n) of the coordinate indices. For n prime, a cyclic code is called a quadratic residue code if, for $0 < b < n$, the coordinate permutation corresponding to the mapping $i \rightarrow bi$ (modulo n) leaves the code invariant if and only if b is a quadratic residue of n . For the rest of this paragraph and in section III we assume that $q = 2$ and n is a prime, which we will refer to by p , such that $p \equiv -1$ modulo 8. These codes were originally defined in a different context by Gleason (1961) and then in this fashion by Prange. Both Gleason (1961) and Prange from their different points of view demonstrated the following facts about quadratic residue codes A of dimension $(p - 1)/2$.

(a) There are exactly two quadratic residue codes of dimension $(p - 1)/2$. These are equivalent and hence have the same weight distributions.

(b) $A_i = 0$ unless i is a multiple of 4.

(c) If B is the orthogonal of A , B is a quadratic residue code of dimension $(p + 1)/2$, and B is obtained from A by adjoining the all one vector. This implies

(d) $B_i = 0$ unless $i = 0$ or 3 modulo 4.

(e) Prange showed that if the minimum weight in B is m , m must be odd, and $m + 1$ is then the minimum weight in A . From this it follows that the maximum weight in A is $n - m$.

II. THE POWER MOMENTS

Let $\gamma = q - 1$. MacWilliams (1962) binomial moments are

$$\sum_{j=0}^n \binom{j}{\nu} A_j = q^{k-\nu} \sum_{j=0}^n (-1)^j B_j \gamma^{\nu-j} \binom{n-j}{n-\nu}, \quad B_1$$

$$\sum_{j=0}^n \binom{n-j}{\nu} A_j = q^{k-\nu} \sum_{j=0}^n \binom{n-j}{n-\nu} B_j, \quad B_2$$

where $\binom{a}{b} = 0$ if $b > a$; ν can take any integral values from 0 to n so that (B_1) (or (B_2)) is a set of $n + 1$ equations.

The power moment identities which can be derived from these are the following:

$$\sum_{j=0}^n (j^r) A_j = \sum_{j=0}^n (-1)^j B_j \left(\sum_{\nu=0}^r \nu! S(r, \nu) q^{k-\nu} \gamma^{\nu-j} \binom{n-j}{n-\nu} \right), \quad P_1$$

$$\sum_{j=0}^n (n-j)^r A_j = \sum_{j=0}^n B_j \left(\sum_{\nu=0}^r \nu! S(r, \nu) q^{k-\nu} \binom{n-j}{n-\nu} \right). \quad P_2$$

Here $S(r, \nu)$ is a Stirling number of the second kind,

$$S(r, \nu) = \left[\frac{\Delta^j x^r}{\nu!} \right]_{x=0} = \frac{1}{\nu!} \sum_{i=1}^{\nu} (-1)^{\nu-i} \binom{\nu}{i} i^r.$$

We get an infinite system of equations for either P_1 or P_2 as r can take any positive values or zero. P_1 and P_2 are valid for $r > n$ provided we understand that $\binom{a}{b} = 0$ if $b < 0$. If $r < n$, then $\sum_{j=0}^n$ on the right sides of P_1 and P_2 can be replaced by $\sum_{j=0}^r$.

We will give the derivation of P_2 from B_2 . P_1 follows from B_1 similarly. From Jordan (1950) we know that

$$(n-j)^r = \sum_{\nu=0}^r \nu! \binom{n-j}{\nu} S(r, \nu)$$

so that

$$\begin{aligned} \sum_{j=0}^n (n-j)^r A_j &= \sum_{j=0}^n \left(\sum_{\nu=0}^r \nu! \binom{n-j}{\nu} S(r, \nu) \right) A_j \\ &= \sum_{\nu=0}^r \nu! S(r, \nu) \left(\sum_{j=0}^n \binom{n-j}{\nu} A_j \right) \\ &= \sum_{\nu=0}^r \nu! S(r, \nu) \left(q^{k-\nu} \sum_{j=0}^n \binom{n-j}{n-\nu} B_j \right) \quad \{\text{by } B_2\} \\ &= \sum_{j=0}^n B_j \left(\sum_{\nu=0}^r \nu! S(r, \nu) q^{k-\nu} \binom{n-j}{n-\nu} \right). \end{aligned}$$

It can also be shown that the binomial identities can be derived from the power moment identities.

Since we are interested in solutions for codes, from now on we assume $A_0 = B_0 = 1$.

The following is a list of the first few equations in P_1 for $q = 2$.

$$r = 0 \quad \sum A_j = 2^k, \quad (1)$$

$$r = 1 \quad \sum j A_j = 2^{k-1} n - 2^{k-1} B_1, \quad (2)$$

$$r = 2 \quad \sum j^2 A_j = 2^{k-2} n (n + 1) - 2! 2^{k-2} n B_1 + 2! 2^{k-2} B_2, \quad (3)$$

$$\begin{aligned} r = 3 \quad \sum j^3 A_j &= 2^{k-3} (n^3 + 3n^2) \\ &\quad - 2^{k-3} (3n^2 + 3n - 2) B_1 + 3! 2^{k-3} n B_2 \\ &\quad - 3! 2^{k-3} B_3, \end{aligned} \quad (4)$$

$$\begin{aligned} r = 4 \quad \sum j^4 A_j &= 2^{k-4} (n^4 + 6n^3 + 3n^2 - 2n) \\ &\quad - 2^{k-2} (n^3 + 3n^2 - 9n + 7) B_1 \\ &\quad + 2^{k-2} (3n^2 + 3n - 4) B_2 - 4! 2^{k-4} n B_3 \\ &\quad + 4! 2^{k-4} B_4. \end{aligned} \quad (5)$$

The Assmus-Mattson (1961) result on the sum of the squares of the weights in a code can be stated as follows. Consider the matrix M whose rows are all the elements in A over a general $GF(q)$, q a prime. Assmus and Mattson assume no column of M is 0 (this is equivalent to $B_1 = 0$). Let $c_1, \dots, c_s$ be a largest set of columns of M such that no two are multiples of each other. Let j_i be the number of columns of M which are nonzero multiples of c_i . Then

$$B_2 = \gamma \sum_{i=1}^s \binom{j_i}{2}, \quad n = \sum_{i=1}^s j_i.$$

The equation in P_1 for $r = 2$ is

$$\begin{aligned} \sum i^2 A_i &= q^{k-1} \gamma n + q^{k-2} \gamma^2 n (n - 1) + B_2 (q^{k-2} 2) \\ &= q^{k-1} \gamma n + q^{k-2} \gamma^2 n (n - 1) + \gamma q^{k-2} 2 \sum_{i=1}^s \binom{j_i}{2} \\ &= q^{k-1} \gamma n + q^{k-2} \gamma^2 n (n - 1) + \gamma q^{k-2} \left(\sum_{i=1}^s j_i^2 - \sum_{i=1}^s j_i \right) \\ &= q^{k-1} \gamma n - q^{k-2} \gamma^2 n - \gamma q^{k-2} \sum_{i=1}^s j_i + q^{k-2} \gamma^2 n^2 + q^{k-2} \gamma \sum_{i=1}^s j_i^2 \\ &= q^{k-2} \gamma n (q - \gamma - 1) + q^{k-2} \gamma^2 n^2 + q^{k-2} \gamma \sum_{i=1}^s j_i^2 \\ &= \gamma q^{k-2} \left(n^2 \gamma + \sum_{i=1}^s j_i^2 \right) \end{aligned}$$

which is the Assmus-Mattson identity.

III. APPLICATIONS

THEOREM. *A unique solution to P_1 (hence also to B_1, B_2, P_2) exists under the following conditions. Only s A_i 's are unknown and $B_1, B_2, \dots, B_{s-1}$ are known.*

PROOF: The first s equations in P_1 are s equations in s unknowns, whose coefficient matrix is van der Monde. Hence there is a unique solution for the unknown A_i 's in the first s equations and each additional equation brings in exactly one additional B_i which can be uniquely solved for.

As an application of the preceding theorem consider the quadratic residue codes, for $q = 2$, of dimension $(p - 1)/2$, $p \equiv -1$ modulo 8. Let A denote such a code and B denote its orthogonal. Mattson-Solomon (1961) give the following bound on the minimum odd weight m in B , $m^2 - m + 1 \geq p$. By remark (e) we know that the minimum weight in A is $m + 1$. For $p = 7, 23, 31$, or 47 this makes enough B_i 's zero so that the conditions of the theorem are fulfilled and the unique solution is precisely the known weight distribution of these codes. Since Muir and Metzler (1930) give simple expressions for van der Monde determinants and also for determinants which are close to van der Monde, the computation of the weights is not arduous.

TABLE 1
WEIGHT DISTRIBUTIONS FOR SOME QUADRATIC RESIDUE CODES

	7	23	31	47	71
A_0	1	1	1	1	1
A_4	7				
A_8		22×23	15×31		
A_{12}		56×23	280×31	276×47	35×71
A_{16}		11×23	589×31	7590×47	$2,345 \times 71$
A_{20}			168×31	49588×47	$186,186 \times 71$
A_{24}			5×31	81720×47	$4,340,910 \times 71$
A_{28}				35420×47	$37,861,505 \times 71$
A_{32}				3795×47	$129,893,225 \times 71$
A_{36}				92×47	$181,404,764 \times 71$
A_{40}					$103,914,580 \times 71$
A_{44}					$24,093,685 \times 71$
A_{48}					$2,170,455 \times 71$
A_{52}					$71,610 \times 71$
A_{56}					670×71
A_{60}					7×71

As an example consider the $p = 23$ case. $m = 7$ so that the only nonzero A_i possible are A_8 , A_{12} , and A_{16} . The first three equations in P_1 are

$$\begin{aligned} A_8 + A_{12} + A_{16} &= 23 \times 89, \\ 8A_8 + 12A_{12} + 16A_{16} &= 2^{10} \times 23, \\ 8^2A_8 + 12^2A_{12} + 16^2A_{16} &= 2^9 \times 23 \times 24. \end{aligned}$$

Hence $A_8 = 22 \times 23$, $A_{12} = 56 \times 23$, $A_{16} = 11 \times 23$.

When $p = 71$, the general known properties of quadratic residue codes together with the power moment identities do not yield a unique weight distribution for A , but do eliminate all except four possibilities. The correct weight distribution was then determined by examining sets of vectors in the code. All computations were done by hand. This new result is listed along with the cases $p = 7, 23, 31, 47$. These cases had been known previously, the calculation for $p = 47$ being due to Gleason (1961).

ACKNOWLEDGMENTS

I wish to thank John Pierce for aid in solving P_1 and Eugene Prange for helpful suggestions and illuminating discussions.

RECEIVED: September 28, 1962; revised January 16, 1963

REFERENCES

- ASSMUS, E. F., AND MATTSO, H. F. (1961), Error correcting codes: An axiomatic approach. ARM No. 269, Applied Research Laboratory, Sylvania Electronic Systems, Waltham, Mass.
- GLEASON, A. M. (1961-1962), private communications.
- JORDAN, C. (1950), "Calculus of Finite Differences," 2nd ed. Chelsea, New York.
- MACWILLIAMS, F. J. (1962), A theorem on the distribution of weights in a systematic code. *Bell System Tech., J.* **42**, 79-94.
- MATTSO, H. F., AND SOLOMON, G. (1961), A new treatment of Bose-Chaudhuri codes. *J. Soc. Ind. Appl. Math.* **9**, 654-669.
- MUIR, T., AND METZLER, W. (1930), "Theory of Determinants." Privately published, Albany, N. Y.
- PRANGE, E. (1961-1962), private communications.
- ZEILER, N. (1962), A note on the mean square weight for group codes. *Inform. and Control* **5**, 87-89.